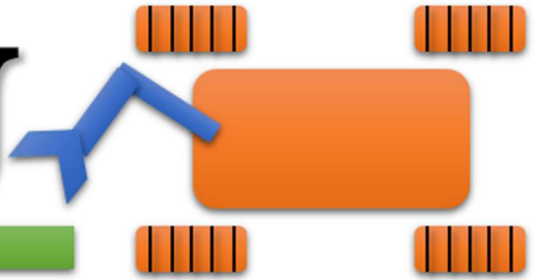


IRoC-U



ISRO Robotics Challenge - URSC

Let's build a space robot

Proposal Report

by

Manik

10624

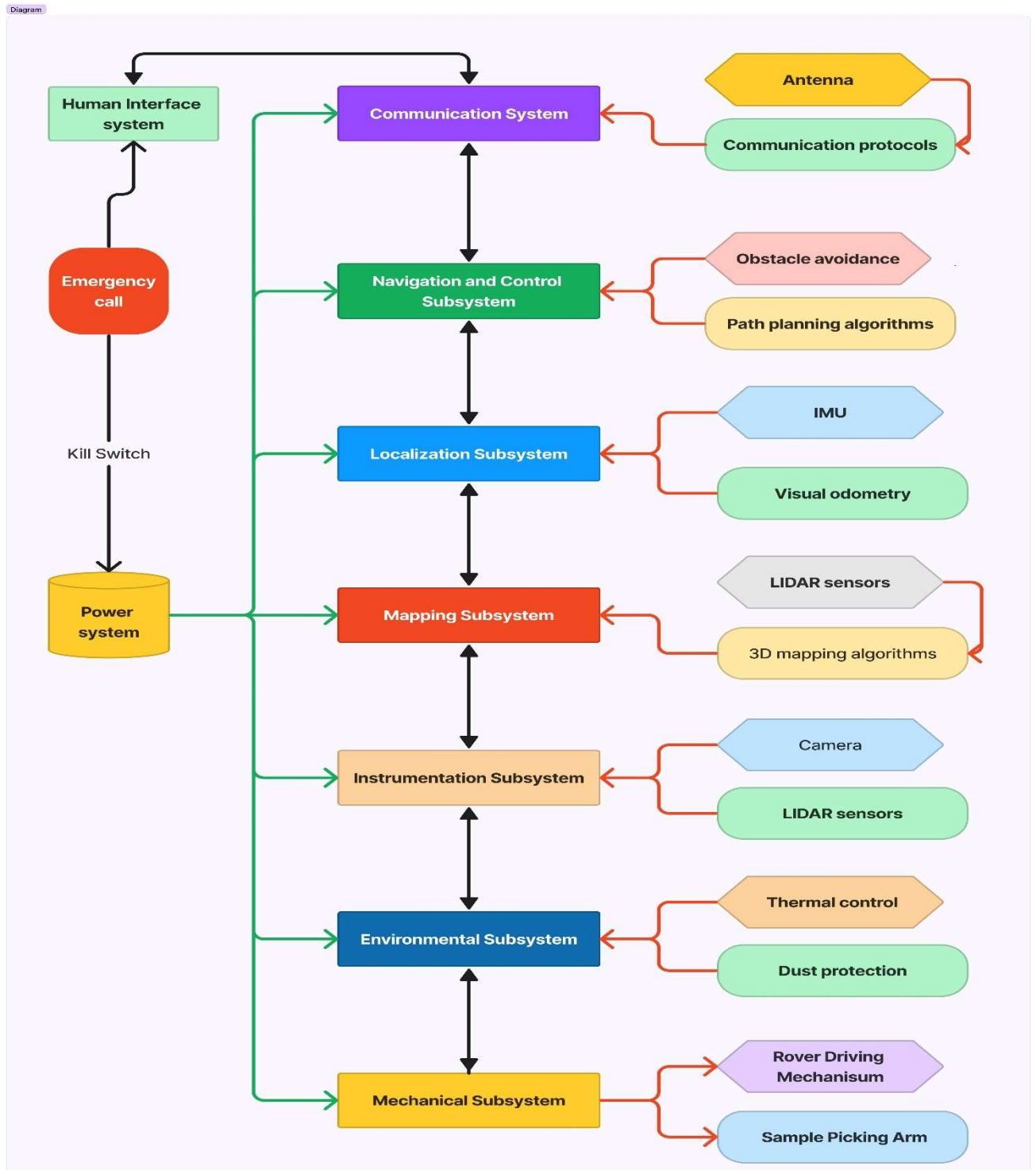
December 2023

**SHRI GURU GOBIND SINGHJI
INSTITUTE OF ENGINEERING
& TECHNOLOGY**

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System Architecture



Communication Subsystem:

Antenna: Facilitates communication with the competition control system.

Communication protocols: Ensure reliable data transfer for receiving commands and sending data during the competition.

Navigation and Control Subsystem:

Path planning algorithms: Plan a route to the goal point based on the constructed map of the arena, considering obstacles and the rover's capabilities.

Obstacle avoidance algorithms: Adjust the planned route to navigate around obstacles detected by sensors in the arena.

Localization Subsystem:

Inertial Measurement Unit (IMU): Provides information about acceleration and angular velocity for position estimation.

Visual odometry: Uses cameras to track features in the arena, estimating changes in position over time.

Mapping Subsystem:
3D mapping algorithms: Construct a map of the competition arena using data from cameras and other sensors.

LIDAR (Light Detection and Ranging): Measures distances to objects in the arena, aiding in terrain mapping.

Instrumentation Subsystem:

Cameras: Capture images for visual odometry and mapping in the arena.

LIDAR sensors: Measure distances to create detailed terrain maps specific to the competition arena.

Environmental Subsystem:

Thermal control systems: Regulate internal temperatures based on Earth's conditions.

Dust protection measures: Shield sensitive equipment from arena-like debris or dust.

Mechanical Subsystem:

Chassis: Provides structural support suitable for the arena.

Wheels or tracks: Allow the rover to move across the competition arena.

Articulation mechanisms: Adapt to the terrain of the arena, mimicking moon-like conditions.

Human Interface Subsystem:

User interface: Allows operators to monitor the rover's progress during the competition, receive data, and send high-level commands.

Navigation Process for Rover Competition:

Initial Positioning: The rover determines its initial position using the IMU and visual odometry based on features within the competition arena.

Mapping: As the rover moves within the arena, it continuously builds and updates a 3D map using cameras and LIDAR sensors specific to the competition environment.

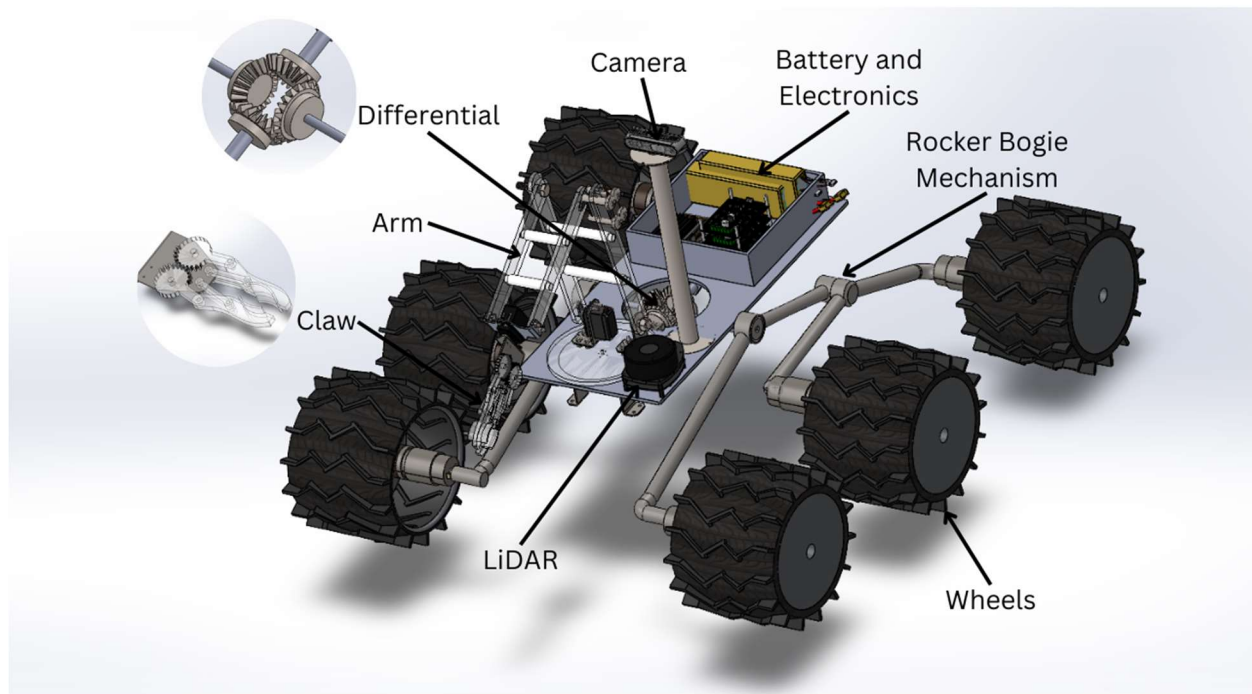
Path Planning: The navigation system plans a route to the goal point within the 12m x 12m arena, considering the arena-specific map, obstacles, and the rover's capabilities.

Obstacle Avoidance: The rover uses sensors to detect arena-specific obstacles in real-time and adjusts its path accordingly.

Communication with Control System: The rover communicates its progress and receives high-level commands from the competition control system, ensuring alignment with competition rules.

Autonomous Movement: The rover autonomously navigates towards the goal point within the competition arena, adapting its route based on real-time data from sensors specific to the arena conditions.

Roving Mechanism



We are using Rocker Bogie as a roving mechanism. This mechanism is more stable and efficient for traversing in our surrounding conditions because of its bogie segment division and basic reason is it contains six wheels which are three wheels on each side. It also contains a differential to increase the stability of the main chassis.

Estimated Weight:36kg.

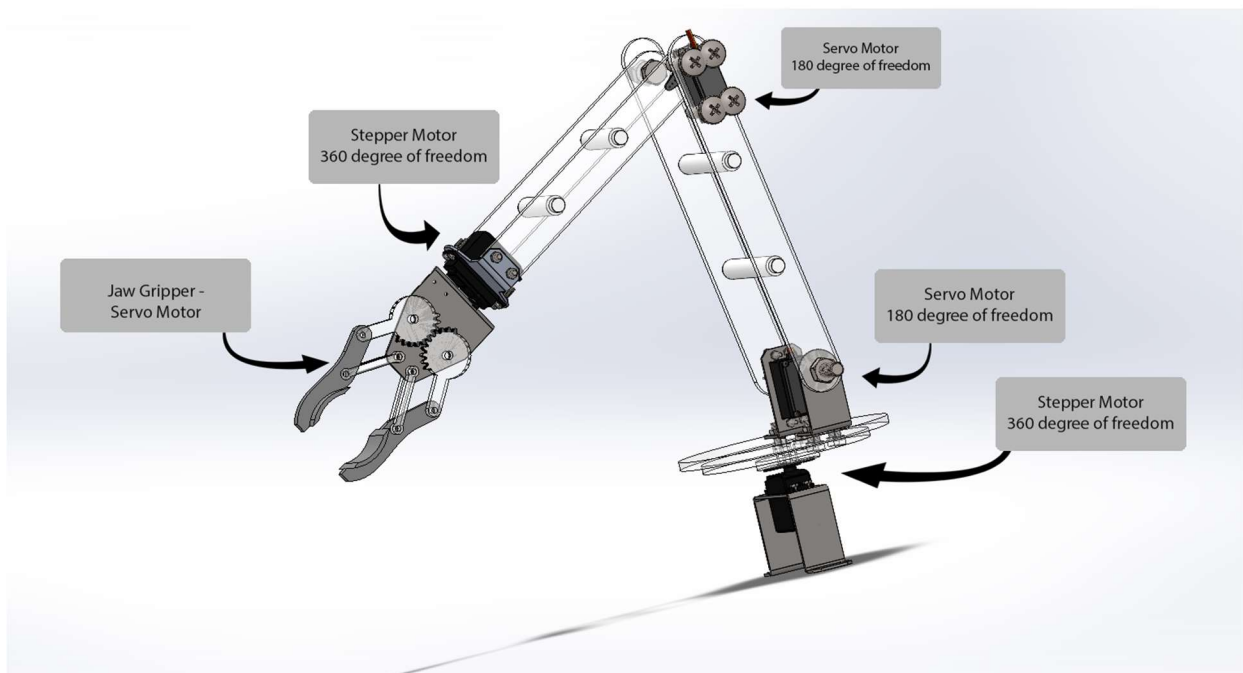
Size of bot (L X b X h): 70X60X65

- **Sub-drives and Sub-integration:** The rocker-bogie contains different sub-drives and sub-integrations such as bogie segments with different pivot points, differential, and swerve drive.
 - a. **Bogie Segmentation:** The individual segments are interconnected with a free pivot point, acting like a suspension system of the mechanism, providing greater stability.
 - b. **Differential (Four-Geared System):** The differential is a four geared system that provides stability to the main chassis by dividing movement from one bogie segment to another, increasing contact between wheels and ground.
 - c. **Swerve Drive:** A swerve drive is used to provide free movement to wheels in all axes.

JUSTIFICATION FOR THE USED MECHANISM:

1. **Stability:** The six-wheel design and the differential system ensure that the vehicle remains stable, even when traversing over uneven terrain. The differential also helps to distribute the load evenly across all wheels, reducing the risk of tipping over.
2. **Mobility:** The swerve drive allows for free movement of the wheels in all axes, enhancing the vehicle's movements. This is particularly useful in navigating around obstacles or through tight spaces.
3. **Shock Absorption:** The interconnected segments with free pivot points act like a suspension system, absorbing shocks from the terrain and providing a smoother ride.
4. **Efficiency:** The Rocker Bogie mechanism is efficient in terms of energy usage, as it allows for smooth movement with minimal resistance.

Pick and Place Mechanism:



- Utilizes a **Robotic Arm** with six degrees of freedom for precise maneuverability.
- Equipped with a claw mechanism featuring dual grippers for versatile and secure object handling.
- **Robotic Arm:**
 - Provides six degrees of freedom with six-axis movement.
 - Joints offer various degrees of freedom: base joint (360°), middle joints (180° each), and wrist joint (360°).

- Different types of motors used for different joints: stepper motors for the base joint and servo motors for all other joints.
- **Gripper:**
 - Used for picking objects, powered by a servo motor.
 - Features two interlocked gears that move in opposite directions (one clockwise, the other anti-clockwise) to enable object pick-up.

Justification of used Mechanism:

Four Degrees of Freedom: The robotic arm has four degrees of freedom, allowing it to move in a limited number of directions and around certain axes. This makes it simpler and potentially faster than a 6 Degrees of Freedom mechanism, while still being capable of handling a variety of tasks.

Precise Movement: Different types of motors are used for different joints, including a stepper motor for the base joint and servo motors for all other joints. This allows for precise control over the arm's movements, which is crucial for tasks that require a high level of accuracy.

Versatile Gripper: The gripper's design features two interlocked gears moving in opposite directions, enabling it to securely pick up a wide range of objects. This is particularly useful in scenarios where the objects to be moved vary in size, shape, or weight.

Dual Grippers: The presence of dual grippers increases the mechanism's object-handling capabilities. It can pick up two objects at once, improving efficiency, or handle a single object with both grippers for added stability.

Emergency Response System

Mechanical failure: This failure could happen due to damage in important mechanical parts of the rover due to obstacles present in the arena or any other unexpected situations.

Obstacle/Path recognition failure: This problem could happen due to sensor/electronic components failure (i.e. Camera, Lidar, etc.) or due to software-related failures.

Power supply failure: This problem may arise due to battery-related problems or due to inefficient connections of the components.

Communication failure: This problem could occur due to issues in the communication component or due to signal interference.

Software-related failure: This could happen due to an unexpected error in software or a mismatch in the synchronization of software and hardware.

Failure in the stability of the rover: This failure could happen due to M sand present in the arena or obstacles in the arena.

Description of the Response System: Our Emergency Response System is designed to handle different emergencies effectively.

Mechanical Failure

Approach: Continuous monitoring and visual inspections.

Method: Physical inspection by experienced team members.

Reasoning: Direct observation allows the detection of subtle issues not easily captured by automated systems.

Obstacle/Path Recognition Failure

Approach: Utilization of advanced sensors (LIDAR, cameras) for real-time obstacle detection.

Method: Autonomous navigation around obstacles based on sensor data.

Reasoning: Enhances rover autonomy and avoids potential collisions.

Power Supply Failure

Approach: Manual switchover to alternate power sources.

Method: Manually switching to an alternate power source in case of primary power failure.

Reasoning: Ensures uninterrupted rover operations and data collection.

Communication Failure

Approach: Automatically system gets restarted.

Method: Automatic switch to the previous channel in the event of a primary failure.

Reasoning: Ensuring continuous communication for efficient data collection.

Software Related Failure

Approach: Elimination of software glitches and implementation of error-handling mechanisms.

Method: Identification and isolation of software glitches by experienced team members.

Reasoning: Ensuring the software can handle unexpected errors and enable adjustments according to conditions.

Failure in the Stability of the Rover

Approach: Passing through all conditions from which the rover can be unstable.

Method: Using the rocker bogie mechanism to pass over uneven surfaces, including M sand.

Reasoning: The rocker-bogie mechanism provides efficient and stable rover movement, enhancing overall

stability.

Alternate Emergency Response Systems

To augment mission adaptability, alternative emergency response systems are identified for each scenario:

Obstacle Interference: Implementation of AI-based obstacle detection and machine learning for adaptive navigation.

Extreme Environmental Conditions: Deployable protective measures.

Reason for Choosing the Proposed Emergency Response System: The Emergency Response System is influenced by space mission technologies, aiming to be like successful missions. We focus on being adaptable, and independent, and having backup plans to make sure the rover is reliable and successful.

Hardware Identification

No.	Hardware Details	Subsystem	Category	Quantity Need	Specifications	Estimated cost in INR	Justification for Chosen Type
1	Motor	Roving Mechanism	Planetary Gear motors	6	12v ,96rpm, 121.7 N-cm	1340*6=8040	Ease of control Availability
2	Motor	Roving Mechanism	Servo Motors	4	Metal gear Stall torque = 18 kg-cm Operating voltage = 4.8v~6.6v	1185*4=4740	Ease of control Direction
3	Motor	Claw Mechanism	Stepper motor	1	Step angle = 1.8 Toque = 7.2 kg-cm	1299	Direction control of claw
4	Motor	Claw Mechanism	Servo motors	4	Stall torque = 13 kg-cm Operating voltage = 4.8v~7.2v	320*4=1280	Grabbing and Controlling Of claw
5	Computer	Controlling And communication Mechanism	NVIDIA Jetson nano	1	Micro-USB:5V 2A DC power128-core NVIDIA Maxwell™ GPU Quad-core ARM® A57 CPU 4 GB 64-bit LPDDR4	18,106	Controlling action, Processing sensors data And communication
6	IMU Device	Navigation Mechanism	MPU6050	1	Operating voltage =3-5V DC Gyro Range: $\pm 250, 500, 1000, 2000$ °/s Accelerometer Range: $\pm 2 \pm 4 \pm 8 \pm 16g$	115	3-axis motion tracking, Orientation, Acceleration And direction of Rover
7	Battery	Power system	LiPo battery	1	11.1V 1000mAh 35C 3S Lithium Polymer	8999	Main Power supply Of Rover
8	Camera	Sensor and	OAK-D	1	Auto-Focus, Stereo Depth, Integrated 12.3 MP	12367	Collecting data of

		Object detection Mechanism	lite		Display resolution -pixels 4208 x 3120 USB 3.1		the Surrounding Environment
9	Motor Driver	Roving Mechanism	MDD10A	2	Bi-directional control of 2 brushed DC motors. Support motor voltage ranges from 5V to 25V. Maximum current up to 10A continuous and 30A peak (10 seconds) for each channel.	2036*3=6108	Controlling motion and The direction of planetary motor by feeding proper Current and Voltage
10	Switch	Power system	Kill switch	1	Rated Insulation Voltage: 600V Agreed Thermal Current :10A Switch Top Diameter: 39 mm Insulation Resistance: $\geq 60M\Omega$ Contact Resistance: $\leq 25m\Omega$	170	Emergency shutdown
11	Servo motor driver	Claw and Roving Mechanism	SC08A	2	Resolution: 8000 steps = 0.25us. UART: 9600 baud rates. Servo pulse range: 0.5ms to 2.5ms. 8 channels: Servo-driven independently Extendable to 16 Channels: Two controllers	1354*2=2708	Controlling action and Position of servo motors of claw and rover According to input given by computer
12	Stepper motor driver	Claw Mechanism	A4988 Drivers	2	8-35 VDC. Current: 2.0A logic input: 3 – 5.5V	140	Controlling action and Position of stepper motors of claw according to input given the computer
13	Stepper Shield for	Claw Mechanism	CNC shield1	1	Latest Arduino CNC Shield Version3.10	114	compatible Support 4 A4988

	Arduino				4-Axis support GRBL 0.9		drivers for 4 steppers
14	Microcontroller	Claw mechanism	Arduino mega	1	54 Digital I/O terminals 16 Analog Inputs. 4 UARTs 16 MHz crystal clock Operating voltage: 6 ~ 12v.	1402	Controlling claw mechanism by taking input from computer
15	Communication module	Communication system	TS832 transmitter	2	Transmitter 600mW/48CH >up to 1km Frequency control: built-in frequency and phase lock loop. Supply Voltage: 7-24V Current: 220 mA	1249	Video transmission and Communication of Rover
16	Communication module	Communication system	RC832 Receiver	1	Operating voltage =12V Current = 200Ma Control Range: Approx 5 KM	2089	Good receiver for TS832 transmitter. Video transmission and Communication of Rover
17	Communication module	Communication system	NRF24L01	2	Voltage = 3-3.6V, Operating range: 1Km	165*2 =330	Wireless communication for data transfer of rover.
18	LIDAR	Sensor and Object detection Mechanism	YDLIDAR	1	Operating voltage =3.3v Current =330-500mA canning Frequency: 7Hz Ranging Frequency: 3000Hz Distance Range: 0.1-8m based on the Based-on reflectivity of objects.	8710	Distance sensing
19	Chassis material	Chassis	Aluminum Steel alloy	15	Profile Size: 2020 Steel alloy Length: 500mm Material: Aluminium Alloy 6063	2985	Lightweight and strong material and easy constructible

20	Wheels	Roving Mechanism	3D printed wheels	6	Abs Filament material, Radius= 8cm	1598	Lightweight and strong material, Proper movements of the rover.
TOTAL BoM						Rs 82477	

Software Identification

Software Being Used for Performing Tasks: Operating System: Linux (Ubuntu 22.04)

a) Robot Operating System (ROS2):

All the software tasks will be formed under this framework

b) Tools used under ROS2:

Transforms (Tfs): These will help in managing the task of hardware and software linkage and synchronization of coordinates of the sensors and main rover mechanism.

URDF File: This xml file is used for the establishing communication between hardware and software by adding linkage like base-link for main rover mechanism.

Odometry: This tool is used for the velocity control and current posing of the bot according map.

Root-Footprint: This package is used for cost map and locale map generation for path planning.

NAV2: This tool used for the enabling following packages.

- i. "Navigation_params. yaml" file for configuring parameters.
- ii. Local planner plugins (e.g., nav2_dwa_planner).
- iii. Global planner plugins (e.g., nav2_navfn_planner).
- iv. Recovery behaviour plugins.
- v. Controller plugins.

c) OpenCV: We are integrating our path planning task with open cv for object detection.

d) MoveIt: We are using this ROS2 tool for our pick and place task.

Algorithm selection for various computation tasks.

Navigation Algorithm:

Step 1: Initialize the ROS2 system and the navigation stack.

- Step 2: Load the map of the environment into the navigation stack.
- Step 3: Set the initial pose of the robot in the map.
- Step 4: Create a goal pose for the robot to navigate to using the `amcl` package.
- Step 5: Use the local planner plugin to generate a path from the current pose to the goal pose.
- Step 6: Send the path to the robot's controller.
- Step 7: Monitor the robot's progress towards the goal pose.
- Step 8: If the robot reaches the goal pose, stop the navigation stack.

Discovering Unknown Areas and Modifying the Map Algorithm:

- Step 1: Initialize the robot's pose and map using the `amcl` package.
- Step 2: Generate a list of frontiers, which are areas that have not been explored yet, using the `frontier\_exploration` package.
- Step 3: Navigate the robot to the closest frontier using the `move\_base` package.
- Step 4: Collect data about the environment using the robot's sensors as it moves.
- Step 5: Build a new map of the environment using the data collected by the sensors using the `gmapping` package.
- Step 6: Merge the new map with the existing map using the `map\_server` package.
- Step 7: Repeat steps 2-6 until all of the unknown areas have been explored and mapped.

Object Picking Algorithm:

- Step 1: Use the robot's sensors to detect objects in the environment.
- Step 2: Identify the object that needs to be picked up.
- Step 3: Generate a path for the robot to reach the object.
- Step 4: Use the robot's arm and gripper to pick up the object.
- Step 5: Keep the object in the desired storage on the rover.

Returning to the Initial Position Algorithm:

- Step 1: Get the robot's current pose using the `amcl` package.
- Step 2: Generate a path from the current pose to the initial pose using the `move\_base` package.
- Step 3: Send the path to the robot's controller.
- Step 4: Monitor the robot's progress towards the initial pose.
- Step 5: Stop the navigation stack if the robot reaches the initial pose.

Hardware And Software Realization Plan

HARDWARE REALIZATION:

No.	Hardware Details	Procurement Source (Market/Fabrication/3D Printing.)	Specifications/ Realization Plan	Quantity	Estimated Cost (INR)
1	Planetary gear motor	Robu.in	Operating voltage = 12 v Speed = 96 rpm	6	8040

			Current=2.15A Torque = 121.7 N-cm		
2	MDD10A Motor Driver	Robu.in	Dual channel Operating voltage = 5~25 v Current = 10 A	3	6108
3	Aluminium steel alloy	Robu.in	Profile Size: 2020 Steel alloy Length: 500mm Mate- rial: Aluminium Alloy 6063	15	2985
4	MG958 Digital High Torque Metal Gear Servo Motor	Robu.in	Stall Torque = 18 kg-cm Operating voltage = 4.8~6.6 V Current = 170 mA Speed = 0.18sec/60°	4	4740
5	MG995 Metal Gear Servo Motor (180° Rotation)	Robu.in	Stall torque = 13 kg-cm Operating voltage = 4.87.2V Speed=20sec/60°	4	1280
6	SC08A SERVO DRIVER	Robu.in	8 channels: Servo-driven independently Extendable to 16 Channels: Two controllers UART: 9600 baud rates Servo pulse range: 0.5ms to 2.5ms Resolution: 8000 steps = 0.25us Operating voltage =7 to 25 V Current = 4A	2	2708
7	NEMA23 Stepper Motor	Robu.in	Step angle = 1.8 Toque = 7.2 kg-cm Current = 1A Operating voltage = 6.6 V	1	1299
8	CNC shield V3	Robu.in	Latest Arduino CNC Shield Version 3.10 4-Axis support GRBL 0.9 compatible Support 4 A4988 drivers for 4 steppers	1	114

9	A4988 driver Stepper Motor Driver	Robu.in	Supply Voltage: 8-35 VDC. Current: 1.0A logic input: 3 – 5.5V.	2	140
10	Arduino Mega	Robu. in	54 Digital I/O terminals 16 Analog Inputs. 4 UARTs 16 MHz crystal clock Operating voltage: 6 ~ 12v.	1	1402
11	NVIDIA Jetson Nano Developer Kit- B01	Robu.in	Micro-USB: 5V 2A DC power adapter: 5V 4A. 128-core NVIDIA Maxwell™ GPU Quad-core ARM® A57 CPU 4 GB 64-bit LPDDR4	1	18106
12	OAK-D lite camera	Luxonix.com	Auto-Focus, Stereo Depth, Integrated 12.3 MP Display resolution -pixels 4208 x 3120 USB 3.1	1	12367
13	TS832 transmitter	Robu.in	Transmitter 600mW/48CH >up to 1km Frequency control: built-in frequency and phase lock loop. Supply Voltage: 7-24 V Current: 220 mA	1	1249
14	RC832 Receiver	Robu.in	Operating voltage =12V Current = 200Ma Control Range: Approx 5 KM	1	2089
15	NRF24L01	Robu.in	Voltage = 3-3.6V, Operating range: 1Km	2	330
16	ORANGE LiPo battery	Robu.in	11.1V 1000mAh 35C 3S Lithium Polymer	1	8999
17	MPU6050	Robu.in	Operating voltage =3-5V DC	1	115

			Current= 3.9 mA 3-Axis Accelerometer and Gyroscope Gyro Range: $\pm 250, 500, 1000, 2000$ °/s Accelerometer Range: $\pm 2 \pm 4 \pm 8 \pm 16g$		
18	Kill switch	Robu.in	Rated Insulation Voltage: 600V Agreed Thermal Current :10A	1	170
19	YDLIDAR X2	Robu.in	Operating voltage =3.3v Current =330-500mA canning Frequency: 7Hz Ranging Frequency: 3000Hz Distance Range: 0.1-8m based on the Based-on reflectivity of objects.	1	8710
20	3D printed wheels	3D printer	Abs Filament material, Radius= 8cm	6	1598
TOTAL BoM				Rs 82477	

SOFTWARE REALIZATION: In our bot we distributed software in two main parts according to tasks. These software tasks are Navigation and Pick and Place. We are going to use ROS2 for integration and path planning of the bot (Rover) which also provide a surrounding map, which we can store for further use. And with the integration of open CV for object detection.

For Pick and Place task we are going to use MoveIt tool from ROS2 framework for axis controlling and motor speed maintaining.

NAVIGATION: The navigation task is depended on to tools that are, first is NAV2 which works under ROS2 integration and another is OpenCV which will be working for object detection on this basis of the features of the objects.

Shortly we are synchronizing these two different software units for navigation task and object traversing. We are going to integrate our scanner hardware to with ROS2 by using the libraries and packages it includes and it will help us to get point cloud of bot surroundings in Rviz, and we will plot this timelapse point cloud on the **SLAM** (Simultaneous Localization and Mapping) which will give us map of the surrounding.

We will store this map and according to the locale map and cost map, we will plan a path to reach the required goal by using the nav2 dynamic window planner. According to this planner, the odometry and amcl will provide commends to the controller and base-link.

Meanwhile, the OpenCV objective is to detect the colour detection and diameter measurement of the craters in map and provide the intel to the path planner. The object detection will based on colour (RGB & HSV), for block avoiding, traversing and sample picking. Another view is to train bot in such a manner that it will detect the crater and its diameter and intel to the planner.

Pick And Place: The integration of OpenCV and MoveIt within the ROS2 framework is used for pick and place. OpenCV is used for the visual perception aspect, focusing on object detection, recognition, and measurement. The system utilizes RGB and HSV colour spaces to facilitate accurate colour detection of objects. MoveIt is used to plan and execute the pick and place actions.

OpenCV, a dedicated node is established to process camera input, perform object detection, and publish relevant information such as object position, colour, and diameter on specific ROS topics. MoveIt is integrated with ROS2 as another node, managing the motion planning and execution tasks. This node subscribes to the object information published by the OpenCV node, incorporating visual feedback into the planning process. MoveIt relies on a detailed description of the robot's kinematics and dynamics, typically represented using the Unified Robot Description Format (URDF). This model includes information about joint limits, joint types, and the physical structure of the robot. MoveIt generates a trajectory that specifies how the robot's joints should move over time to execute the planned motion.

Test Plan

Identification of required tests.

Chassis Testing: The chassis, or the framework of the rover, is tested for its durability and ability to support the whole structure¹. It is typically composed of a metal frame — usually steel or a combination of metals — supported on springs or some other type of suspension system.

Rocker-Bogie Suspension Testing: The Rocker-Bogie suspension system, a key feature in many rovers including the Mars rovers, is tested for its ability to navigate over rough terrain and obstacles².

Wheel Testing: The wheels of the rover are tested for their durability and ability to traverse different types of terrain.

Mechanical System Testing: Any mechanical systems, such as robotic arms or drilling mechanisms, are tested for their functionality and durability.

Vibration Testing: The rover is tested for its ability to withstand the intense vibrations during the launch and landing phases of the mission.

Power Systems Testing: The rover's power systems, including batteries, are tested for their efficiency and reliability.

Communication Systems Testing: The rover's communication systems are tested to ensure they can communicate with the mission control centre and other components of the rover.

Sensor and Instrument Testing: The rover's electronic sensors and instruments are tested for their functionality and accuracy.

Software and Firmware Testing: The software and firmware that control the rover's systems are tested to ensure they function correctly and can handle potential errors.

Circuit Testing: The rover's electronic circuits are tested for their functionality and durability.

Unit Testing: This involves testing individual components of the software to ensure that they function correctly.

Integration Testing: This test checks if different software modules work correctly when integrated.

We are dividing our test plans in two parts, which are inter-software connection and software-hardware integration. These are the two main integrations that should be checked by using our test plans.

1. TEST PLAN FOR INTER-SOFTWARE CONNECTIONS:

Inter-linkage between two nodes and tools is very important in ROS2 because without these interconnections, they will not be able to communicate with each other and communication between them is very important for data transfer from one tool or node to another. To test this communication among them in ROS2 framework we are using another ROS2 tool named rqt graph as a test plan. This rqt graph is an inbuilt tool of the ROS2 which provide us the graph, about the interconnection established for the communication. This graph gives us idea about the how intel is transferred and what is the flow of it. Rqt graph gives us the idea about how many tools or nodes are currently working and their interconnections.

2. TEST PLAN FOR SOFTWARE AND HARDWARE INTEGRATION:

Hardware and software synchronization is very important for smooth working of overall bot. Hardware and software synchronization directions that, all the components we are using in the bot should perfected connected with the software (ROS2) that there will not be any delay between the command send by the software and execution this command on the hardware and another view point is to intact integration hardware to the software. For this synchronization we are taking test plan according to the components used in hardware.

1. Test plan for Oak-D lite:

Oak-D lite is camera and depth sensing module used for path planning task, for the precise working of this integration we will give every time different fps frame from camera and we will take a look how it behaves. Another way is to give dynamic bodies and surroundings to observe the sensing, display of point cloud on the Rviz, its localization on map and response to hardware. Observing this much information is a dynamic state test plan for this component.

2. To locate object positions using computer vision and perform arm manipulation to reach the given positions or joint angles using the Moveit framework.

Python script detects the object using camera data, publishes TF, and subscribes to camera topics (/camera/color/image_raw, /camera/aligned_depth_to_color/image_raw). It finds object pose and sends transform relative to base_link. Arm movement sequence: P1 → D with positions: P1 [0.35, 0.1, 0.68], D [-0.37, 0.12, 0.397]. MoveIt Servo streams EEF velocity commands for concurrent execution.

System Specifications

System Type	Components	Functionality
Power System	Battery Cells	Battery cells: Store electrical energy generated by the rover's power source.
Navigation System	Camera, Sensors	Cameras: Capture high-resolution images of the Martian terrain for autonomous navigation and mapping purposes. Sensors: Collect data related to the rover's position, orientation, and surrounding environment for navigation purposes
Thermal Control system	Insulation and heaters	Insulation: Protects sensitive electronics from extreme temperatures, ensuring their optimal functionality. Heaters: Prevents critical components from getting too cold during cold Martian nights, maintaining operational temperatures. Radiators: Dissipate excess heat generated by the rover, preventing overheating.

Computer and Electronics	Onboard microprocessors, electronic components	Onboard microprocessors: Process and manage data collected by various scientific instruments and control systems. Electronic components: Control the operation of scientific instruments, ensuring proper functionality and data acquisition.
Pick And Place	Robotic Arm(4-axis)	Robotic arm: A versatile mechanical limb equipped with gripper that collects the pipe from the surface and place to the cylinder.

Project Management

Initially we have divided the major task into minor task so that it can be easily achieve. We have chosen the appropriate sensors which are required to complete the task. Which includes cameras, which can capture visual images for object detection, lane detection, recognition of obstacles; lidar which measure the distance and angle of objects for obstacle avoidance, mapping, and localization.

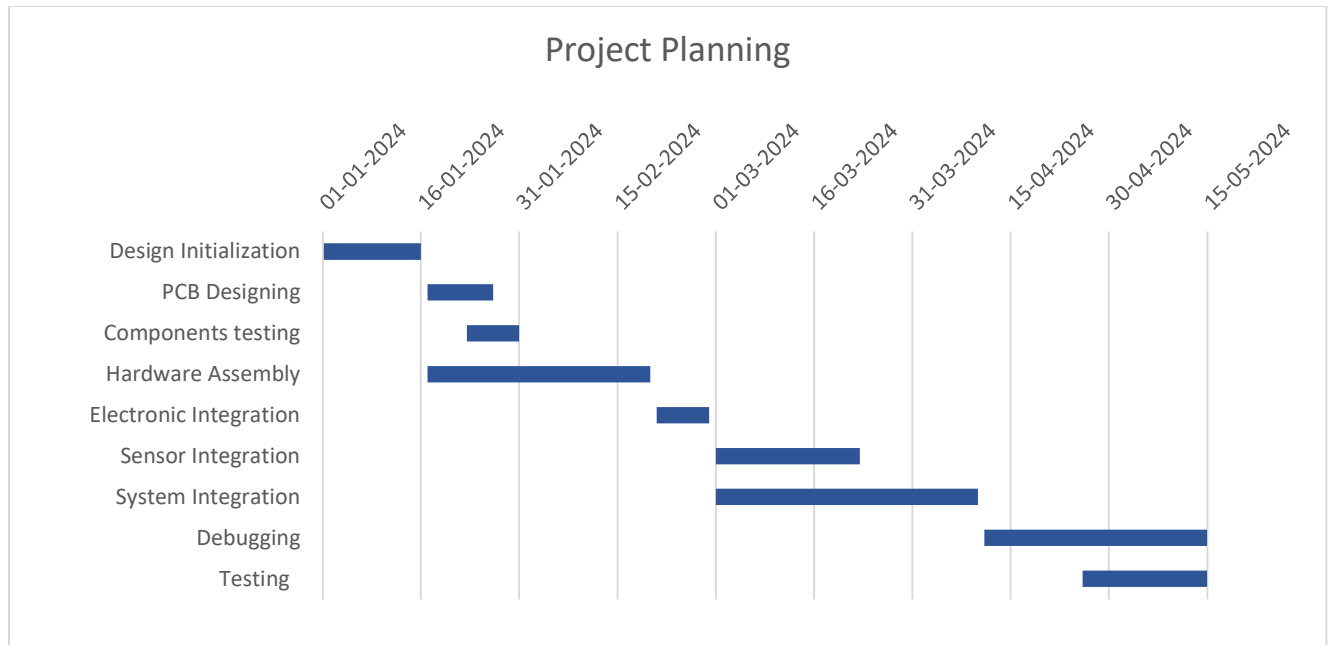
Once we understood the problem we were trying to solve, we began exploring various solutions. We first experimented

with four-wheel mechanisms, both with and without conveyors. However, neither design delivered the desired performance. Seeking inspiration, we turned to online sources and discovered the rocker-bogie mechanism. Intrigued by its potential for stability, we tested it in both differential and non-differential configurations. Ultimately, the differential setup emerged as the clear winner, significantly improving overall stability, particularly for the upper body.

Next, we tackled the challenge of manipulating pipes. We evaluated different claw designs, each offering unique advantages and drawbacks. Now, with a clearer understanding of the individual tasks involved, we've assigned roles based on each team member's expertise. The mechanical team is utilizing CAD software to determine the centre of mass, calculate torque requirements for movement, and analyse performance on inclines of up to 15 degrees. The electronics team is focusing on selecting the

No.	Task	Main Responsibility	Deadline for Completion	Estimated Cost	Secondary Responsibility
1.	Mechanical Design	All Member	25-12-2023	30000	Not Required
2.	CAD Design	Shubham	12-01-2024	00	Not Required
3.	Electrical systems (PCB)	Nikhil	17-02-2024	10000	Electronics team
4.	Microcontroller and sensors	Utkarsh	17-02-2024	30000	Electronics team
5.	Communication Systems	Gaurav	17-02-2024	10000	Electronics team
6.	Testing and Quality Assurance	Prajwal	27-02-2024	4750	All Members
7.	Rover Control Software	Abhishek	25-12-2024	00	Software Team
8.	Autonomous Navigation	Rushikesh	02-03-2024	00	Software Team
9.	Server and Client	Tanmay	01-03-2024	00	Software Team
10.	Open CV	Shantanu	02-02-2024	00	Software Team
11.	Assembly and Integration	All Team Members	25-04-2024	10000	All Members
12.	Documentation and Reporting	Utkarsh	15-01-2024	00	All Members
Total				94,750	

microprocessor, microcontroller, and sensors needed for obstacle avoidance and feedback. Finally, the software team is investigating the best technologies for automating rover operations and ensuring successful mission completion.



Novelty in the overall proposal

1. **Arm:** 360-degree claw rotation, irrespective of size and orientation of object claw able to grab.
2. **Computer:** GPU and CPU integration are different also GPU helps to perform ros2 graphics works effectively. Video transmission enhancement. Versatile, allowing for easy programming and integration of various sensors and modules.
3. **Advanced Sensors:**
 - Lidar:** For accurate mapping and obstacle avoidance.
 - OAK-D Lite (with stereo camera):** Implement computer vision for object detection and navigation. At this price point, it provides better specifications.
4. **Human Interface:** Data given from LIDAR, IMU, and Camera of Telemetry will be shown on the Human interface
5. **Custom PCBs:** Printed circuit boards (PCBs) for the power system and control system which provide efficiency and stable for the electronics of the rover.
6. **Wireless Communication:** We are using a modular approach for communication

- I. **Video Transmission System:** Facilitates real-time visual communication by streaming live video from the rover's camera to a control station. Enables remote exploration through a wireless link, providing a direct visual perspective of the rover's surroundings.
- II. **Data Transfer System:** Ensures seamless exchange of critical information from the rover to the control station. Transmits data on the rover's current location, temperature, and detected objects, enabling comprehensive monitoring and decision-making during the mission.

7. **Advanced Technologies:** To control pick and place mechanism using MoveIt framework and for navigation purpose using NAV2 and Slam mapping. All above tools are integrated with Ros2. For object detection we are using OpenCV.

8. **3D Printing:** Utilize 3D printing for custom components and enclosures like claw, other pick and place mechanisms, and stands for Camera.

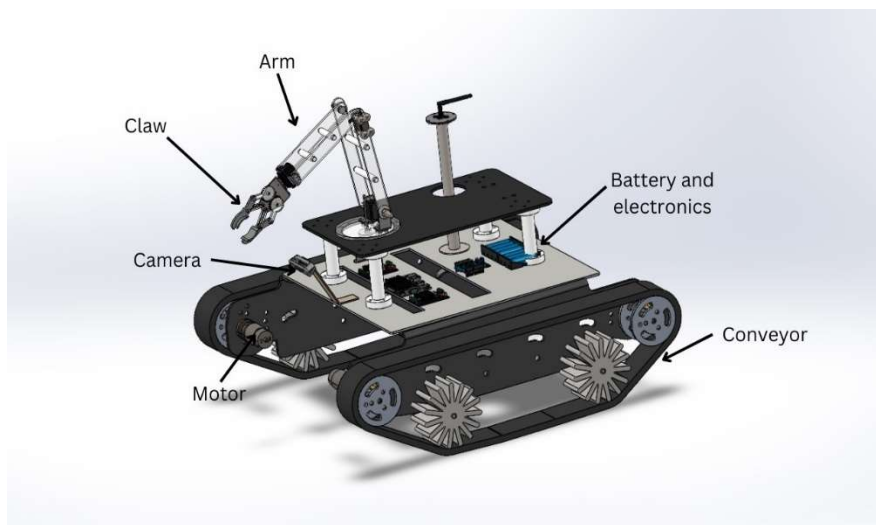
9. **Rover chassis:** Light in weight and strong Chassis made of aluminium steel alloy because of structure of aluminium.

10. **Simulation:** For error minimization we are providing an arena and bot simulation on the ros2. This will be represented on the Gazebo for demo working purpose.

11. **Specialized Motors:** Special planetary gear of high torque and low speed for Roving mechanism. Specialize Stepper and servo of high torque for pick and place mechanism which can easily grab the object with perfection.

Other Relevant Information

Following an extensive design, testing, and analysis process, we are pleased to present the initial robotic chassis, depicted in the attached CAD image.



After a thorough brainstorming session and rigorous testing, we have determined that the rocker bogie mechanism stands out as the optimal choice for navigation. This mechanism ensures stability across diverse terrains during arena maneuvering. The stability it provides is pivotal for securing our camera, lidar, and other sensors, enhancing their ability to capture precise and reliable navigation data. This strategic design decision significantly improves the overall performance and reliability of our robotic system.

Initial level PCB design:

